

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 1

MH1100 Calculus I

Homework 1, 15 Sept.

Due by 6:00 PM on September 21, 2023. Your score will have a cap of 10 marks.

Question 1

(1 marks)

Discuss the behavior of the following functions, explaining why they have no limit as $x \rightarrow 0$.

$$f(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases} \quad g(x) = \begin{cases} \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases} \quad h(x) = \begin{cases} 0, & x \leq 0 \\ \sin\left(\frac{1}{x}\right), & x > 0 \end{cases}$$

[Solution:]

- (a) We can easily obtain that $\lim_{x \rightarrow 0^-} f(x) = 0$ and $\lim_{x \rightarrow 0^+} f(x) = 1$. The two one-sided limits exist but are not equal. $f(x)$ has no limit as $x \rightarrow 0$. $f(x)$ itself is not defined at $x = 0$.
- (b) $g(0) = 0$ is defined. But $\lim_{x \rightarrow 0^-} g(x) = -\infty$ and $\lim_{x \rightarrow 0^+} g(x) = +\infty$. $g(x)$ has an infinite limit or its limit is not defined as $x \rightarrow 0$.
- (c) $h(0) = 0$ is defined, and $\lim_{x \rightarrow 0^-} h(x) = 0$. But $\lim_{x \rightarrow 0^+} h(x)$ does not exist, as $\sin\left(\frac{1}{x}\right)$ does not approach a single value as $x \rightarrow 0^+$. Thus, $h(x)$ has no limit as $x \rightarrow 0$.

Question 2

(1 marks)

- (a) If $\lim_{x \rightarrow 0} \frac{f(x)-5}{x-2} = 3$, find $\lim_{x \rightarrow 0} f(x)$.
- (b) If $\lim_{x \rightarrow 2} \frac{f(x)-5}{x-2} = 4$, find $\lim_{x \rightarrow 2} f(x)$.

[Solution:]

- (a) When x approaches 0, $x = 2$ can be avoided by restricting the discussion on an open interval $(0 - \delta, 0 + \delta)$ with $\delta < 2$. As a result, we can express $f(x)$ as

$$f(x) = \frac{f(x)-5}{x-2} \cdot (x-2) + 5.$$

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Given $\lim_{x \rightarrow 0} \frac{f(x)-5}{x-2} = 3$, $\lim_{x \rightarrow 0} (x-2) = -2$, and $\lim_{x \rightarrow 0} 5 = 5$, we have

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \left[\frac{f(x)-5}{x-2} \cdot (x-2) + 5 \right] \\ &= \lim_{x \rightarrow 0} \frac{f(x)-5}{x-2} \cdot \lim_{x \rightarrow 0} (x-2) + \lim_{x \rightarrow 0} 5 \\ &= 3 \cdot (-2) + 5 \\ &= -1.\end{aligned}$$

(b) When x is close to 2 but not equal to 2, we can express $f(x)$ as

$$f(x) = \frac{f(x)-5}{x-2} \cdot (x-2) + 5.$$

Given $\lim_{x \rightarrow 2} \frac{f(x)-5}{x-2} = 4$, $\lim_{x \rightarrow 2} (x-2) = 0$, and $\lim_{x \rightarrow 2} 5 = 5$, we have

$$\begin{aligned}\lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} \left[\frac{f(x)-5}{x-2} \cdot (x-2) + 5 \right] \\ &= \lim_{x \rightarrow 2} \frac{f(x)-5}{x-2} \cdot \lim_{x \rightarrow 2} (x-2) + \lim_{x \rightarrow 2} 5 \\ &= 4 \cdot 0 + 5 \\ &= 5.\end{aligned}$$

Question 3

(1 marks)

If $\lim_{x \rightarrow c} [f(x) + g(x)] = a$ and $\lim_{x \rightarrow c} [f(x) - g(x)] = b$, find $\lim_{x \rightarrow c} f(x)g(x)$.

[Solution:] Using the sum law, we have

$$\lim_{x \rightarrow c} [2f(x)] = \lim_{x \rightarrow c} [f(x) + g(x)] + \lim_{x \rightarrow c} [f(x) - g(x)] = a + b.$$

Thus,

$$\lim_{x \rightarrow c} f(x) = \frac{a+b}{2}.$$

Using the difference law, we obtain

$$\lim_{x \rightarrow c} [2g(x)] = \lim_{x \rightarrow c} [f(x) + g(x)] - \lim_{x \rightarrow c} [f(x) - g(x)] = a - b.$$

That is

$$\lim_{x \rightarrow c} g(x) = \frac{a-b}{2}.$$

The product law asserts that

$$\lim_{x \rightarrow c} f(x)g(x) = \lim_{x \rightarrow c} f(x) \lim_{x \rightarrow c} g(x) = \frac{a^2 - b^2}{4}.$$

Question 4

(2 marks)

Evaluate the following limits.

(a) $\lim_{x \rightarrow 0} \frac{\sin 7x}{4x},$

(b) $\lim_{h \rightarrow 0} \frac{\cos h - 1}{h}.$

[Solution:]

(a) In order to use the limit fact that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1,$$

we first rewrite the function by multiplying and dividing by 7:

$$\frac{\sin 7x}{4x} = \frac{7}{4} \left(\frac{\sin 7x}{7x} \right).$$

If we let $\theta = 7x$, then $\theta \rightarrow 0$ as $x \rightarrow 0$, so we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 7x}{4x} &= \frac{7}{4} \lim_{x \rightarrow 0} \left(\frac{\sin 7x}{7x} \right) \\ &= \frac{7}{4} \lim_{\theta \rightarrow 0} \left(\frac{\sin \theta}{\theta} \right) = \frac{7}{4} \cdot 1 = \frac{7}{4}. \end{aligned}$$

(b)

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} &= \lim_{h \rightarrow 0} \frac{(1 - 2 \sin^2 \frac{h}{2}) - 1}{h} \\ &= -2 \lim_{h \rightarrow 0} \frac{\sin^2 \frac{h}{2}}{h} \\ &= - \lim_{h \rightarrow 0} \frac{\sin \frac{h}{2}}{\frac{h}{2}} \sin \frac{h}{2} \\ &= - \lim_{h \rightarrow 0} \frac{\sin \frac{h}{2}}{\frac{h}{2}} \cdot \lim_{h \rightarrow 0} \frac{h}{2} = -1 \times 0 \\ &= 0. \end{aligned}$$

Question 5

(2 marks)

Use the definition to prove that

(a) $\lim_{x \rightarrow a} \sin x = \sin a,$

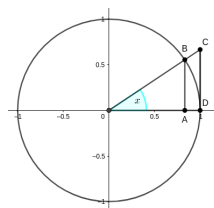
(b) $\lim_{x \rightarrow a} \cos x = \cos a.$

[Solution:]

(a) We first prove $\lim_{x \rightarrow a} \sin x = \sin a$. Given $\epsilon > 0$, we want to find a $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|\sin x - \sin a| < \epsilon$. We have that

$$|\sin x - \sin a| = \left| 2 \cos \frac{x+a}{2} \sin \frac{x-a}{2} \right| \leq 2 \left| \sin \frac{x-a}{2} \right|.$$

From the figure below we know that $\sin x < x$ when $0 < x < \frac{\pi}{2}$. We require that



$0 < |x - a| < \pi$. Then

$$|\sin x - \sin a| \leq 2 \left| \sin \frac{x-a}{2} \right| = 2 \sin \left| \frac{x-a}{2} \right| < 2 \left| \frac{x-a}{2} \right| = |x-a|.$$

If $|x - a|$ is further less than ϵ , then $|\sin x - \sin a| < \epsilon$. Thus, we can choose δ as $\min\{\epsilon, \pi\}$.

(b) Similarly, based on

$$|\cos x - \cos a| = \left| 2 \sin \frac{x+a}{2} \sin \frac{x-a}{2} \right| \leq 2 \left| \sin \frac{x-a}{2} \right|,$$

we can choose δ as $\min\{\epsilon, \pi\}$ to ensure that if $0 < |x - a| < \delta$, then $|\cos x - \cos a| < \epsilon$.

Question 6

(1 marks)

Let $\lim_{x \rightarrow a} f(x) = L$ and $L \neq 0$. Use the ϵ - δ definition to prove that

$$\lim_{x \rightarrow a} \frac{1}{f(x)} = \frac{1}{L}.$$

[Solution:] Since $\lim_{x \rightarrow a} f(x) = L$, given $\epsilon = |L|/2$, there exist δ_1 , such that if $0 < |x - a| < \delta_1$, then $|f(x) - L| < |L|/2$. So,

$$|f(x)| = |L - (L - f(x))| \geq |L| - |L - f(x)| = |L| - |f(x) - L| > |L|/2.$$

Moreover,

$$\left| \frac{1}{f(x)} - \frac{1}{L} \right| = \left| \frac{L - f(x)}{Lf(x)} \right| < \left| \frac{L - f(x)}{L} \right| \cdot \frac{2}{|L|} = |L - f(x)| \cdot \frac{2}{L^2}.$$

Given $\lim_{x \rightarrow a} f(x) = L$, for any positive number $\frac{L^2}{2}\epsilon$, there exist δ_2 , such that if $0 < |x - a| < \delta_2$, then $|f(x) - L| < \frac{L^2}{2}\epsilon$. If $\delta = \min(\delta_1, \delta_2)$, then when $0 < |x - a| < \delta$, we have

$$\left| \frac{1}{f(x)} - \frac{1}{L} \right| = \left| \frac{L - f(x)}{Lf(x)} \right| < \left| \frac{L - f(x)}{L} \right| \cdot \frac{2}{|L|} = |f(x) - L| \cdot \frac{2}{L^2} < \epsilon.$$

Thus,

$$\lim_{x \rightarrow a} \frac{1}{f(x)} = \frac{1}{L}.$$

Question 7

(2 marks)

Let L be a real number. The function $f(x)$ is defined on the real line as

$$f(x) = \begin{cases} 1 + x^2 \cos \frac{1}{x}, & x < 0; \\ L, & x = 0; \\ \sqrt{x+1}, & x > 0. \end{cases}$$

(a) Use the squeeze theorem to prove that $\lim_{x \rightarrow 0^-} f(x) = 1$.

(b) Find $\lim_{x \rightarrow 0^+} f(x)$.

- (c) Based on your conclusions in parts (a) and (b), can you say anything about the limit $\lim_{x \rightarrow 0} f(x)$?
- (d) Can you say anything about the continuity of $f(x)$ at $x = 0$.

[Solution:]

- (a) Let $f_1(x) = 1 - x^2$ and $f_2(x) = 1 + x^2$. We know that $f_1(x) \leq f(x) \leq f_2(x)$. The one-sided limits of $f_1(x)$ and $f_2(x)$ as x approaches 0 from the left are 1. Using the squeeze theorem, we have

$$\lim_{x \rightarrow 0^-} f(x) = 1.$$

- (b) Based on the Root Law,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \sqrt{x+1} = \sqrt{\lim_{x \rightarrow 0^+} (x+1)} = \sqrt{1} = 1.$$

- (c) Both the left-sided and right-sided limits exist and equal 1. So, the limit $\lim_{x \rightarrow 0} f(x)$ exists and equals 1.
- (d) If $L \neq 1$, then $\lim_{x \rightarrow 0} f(x) \neq f(0)$. That means $f(x)$ is not continuous at $x = 0$. If $L = 1$, then $\lim_{x \rightarrow 0} f(x) = f(0) = 1$. In this case, $f(x)$ is continuous at $x = 0$.

Question 8

(2 marks)

Use the definition of continuity and properties of limits to show that the function is continuous on the given interval.

$$(a) \quad f(x) = 2x + \sqrt{x-3}, \quad [3, \infty) \qquad (b) \quad g(x) = \frac{2x-1}{x+6}, \quad (-\infty, -6)$$

[Solution:]

- (a) For $a > 3$, we have

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (2x + \sqrt{x-3}) && \text{(Sum Law)} \\ &= \lim_{x \rightarrow a} 2x + \lim_{x \rightarrow a} \sqrt{x-3} && \text{(Constant Multiple Law, Simple Function Law, Root Law)} \\ &= 2a + \sqrt{\lim_{x \rightarrow a} (x-3)} && \text{(Difference Law, Simple Function Law)} \\ &= 2a + \sqrt{a-3} \\ &= f(a). \end{aligned}$$

So f is continuous at $x = a$ for every a in $(3, \infty)$. Also, $\lim_{x \rightarrow 3^+} f(x) = 6 = f(3)$, so f is continuous from the right at 3. Thus, f is continuous on $[3, \infty)$.

(b) For $a < -6$, we have

$$\begin{aligned}
 \lim_{x \rightarrow a} g(x) &= \lim_{x \rightarrow a} \frac{2x - 1}{x + 6} && \text{(Quotient Law)} \\
 &= \frac{\lim_{x \rightarrow a} (2x - 1)}{\lim_{x \rightarrow a} (x + 6)} && \text{(Sum Law)} \\
 &= \frac{\lim_{x \rightarrow a} 2x + \lim_{x \rightarrow a} (-1)}{\lim_{x \rightarrow a} (x) + \lim_{x \rightarrow a} 6} && \text{(Constant Multiple Law)} \\
 &= \frac{2 \lim_{x \rightarrow a} x + \lim_{x \rightarrow a} (-1)}{\lim_{x \rightarrow a} (x) + \lim_{x \rightarrow a} 6} && \text{(Simple Function Law)} \\
 &= \frac{2a - 1}{a + 6}. \\
 &= g(a).
 \end{aligned}$$

So g is continuous at $x = a$ for every a in $(-\infty, -6)$; that is, g is continuous on $(-\infty, -6)$.